

Software Requirements Specification

Flood Early Warning and Preparedness Checklist Application Kenya

1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for a Progressive Web Application (PWA) that provides flood early-warning information and an El Niño preparedness checklist to residents, schools, and administrators in Kenya.

1.2 Scope

The system covers: (a) location detection (automatic and manual); (b) flood-zone determination using point-in-polygon analysis; (c) precipitation forecast retrieval and multi-tier risk classification; (d) a 23-item preparedness checklist with persistent state; (e) offline capability and print output. The system is client-side only; no personal data leaves the user's device.

1.3 Intended Users (Personas)

Persona	Description	Primary Needs
Persona A	Resident with low-end phone / intermittent 3G connectivity	Fast load, offline access, simple language, manual location entry
Persona B	Household decision-maker	Clear risk tier, actionable checklist, state saved between visits
Persona C	Administrators	Printable checklist and risk summary for posting/distribution

1.4 Definitions

Term	Definition
FCP	First Contentful Paint - time until first text/image renders
PIP	Point-in-Polygon - geometric test whether a coordinate lies inside a polygon
Risk Tier	Green / Yellow / Orange / Red - classification of flood risk
PWA	Progressive Web App - installable web app with offline capability via service worker
KMD	Kenya Meteorological Department
DRR	Disaster Risk Reduction
GeoJSON	Open standard format for geographic features (used for flood polygons)

2. Overall Description

2.1 Product Perspective

A standalone client-side web application. External dependencies: W3C Geolocation API (browser), Open-Meteo REST API (precipitation forecasts, no API key required), and a versioned GeoJSON flood-boundary dataset published by the DRR team.

2.2 Assumptions and Dependencies

A1. Flood polygon data is available as GeoJSON from KMD/DRR and updated at most quarterly.

A2. Open-Meteo API remains available without authentication; the app must degrade gracefully if it is not (see FR-4.1).

A3. Target users have intermittent connectivity; offline use of the checklist is a primary scenario.

A4. All risk thresholds are configurable via a versioned config file (see FR-4.2).

2.3 Constraints

C1. Client-side only — no user accounts, no server-side storage; location data must not be transmitted (NFR-6).

C2. Must run on low-end devices and KaiOS 2.5 (NFR-5).

C3. English UI ; localization designed-for but out of scope.

3. Functional Requirements

Priority order: Must / Should / Could.

3.1 Location Services

ID	Requirement Statement	Priority	Source / User Need	Acceptance Criteria	Test Case	Verification
FR-1.1	The system shall auto-detect the user's location via the W3C Geolocation API (with permission prompt) and use it for risk lookup.	Must	Persona A & B	Location resolved within 10 s of grant; permission prompt shown if not granted.	TC-GEO-01	Automated integration test with mocked coordinates
FR-1.2	The system shall provide a manual search fallback (county/ward selection) when geolocation is denied, unavailable, or times out (>15 s).	Must	Persona A & B	User can complete location entry in ≤3 taps/clicks.	TC-GEO-02	E2E browser test with blocked permissions

3.2 Risk Assessment

ID	Requirement Statement	Priority	Source / User Need	Acceptance Criteria	Test Case	Verification
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FR-2.1	The system shall determine flood-zone membership using a client-side point-in-polygon check against the boundary dataset.	Must	DRR Guidelines	Verdict matches reference oracle for 100% of test points (inside, outside, boundary).	TC-SPAT-01	Unit test with coordinates inside/outside polygon
FR-2.2	The system shall fetch precipitation forecasts from the Open-Meteo API (48 h window, hourly, mm/day).	Must	WMO / KMD Warnings	Fetches on load; refreshed every 30 min; parse errors handled without crash.	TC-API-01	Integration test mocking Open-Meteo REST response
FR-2.3	The system shall classify risk as Green / Yellow / Orange / Red per a documented decision table (thresholds: <20 / 20–50 / 50–100 / ≥100 mm per 24 h) combining rainfall forecast and flood-zone membership.	Must	KMD Alert Framework	Boundary values (19.9, 20, 50, 100 mm) map to the correct tier.	TC-ALGO-01	Unit test parameterized with rainfall/boundary matrices
FR-2.4	The system shall display the risk tier together with the triggering data (rainfall value, time window, zone status) so the result is explainable.	Should	Usability	Tier badge shows source values on tap.	TC-ALGO-02	UI test

3.3 Preparedness Checklist

ID	Requirement Statement	Priority	Source / User Need	Acceptance Criteria	Test Case	Verification
FR-3.1	The system shall render the 23-item El Niño preparedness checklist grouped by phase (Before / During / After).	Must	El Niño Action Plan	All 23 items render in correct groups; no missing or duplicate IDs.	TC-CHK-01	Visual regression & DOM verification
FR-3.2	The system shall allow each checklist item to be marked complete/incomplete with a single tap.	Must	Persona A & B	Toggle updates instantly (<100 ms perceived) and survives scroll/re-render.	TC-CHK-02	Unit + interaction test
FR-3.3	The system shall persist checklist state in localStorage keyed per checklist item and location, restoring it on session restart.	Must	Persona A, B & C	≥95% state restoration across browser restarts; stale state (>30 days) reset with notice.	TC-STO-01	Session-restart browser test
FR-3.4	The system shall provide a print-optimized layout (@media print) rendering checklist and risk summary on ≤2 A4 pages.	Must	Persona C (Schools/Admin)	No clipped content; readable at 100% scale; grayscale-safe.	TC-PRN-01	CSS @media print layout validation

3.4 Reliability & Configuration

ID	Requirement Statement	Priority	Source / User Need	Acceptance Criteria	Test Case	Verification
FR-4.1	The system shall handle API failure gracefully: show cached forecast (with stale flag) or a clear "data unavailable" state with retry.	Must	Reliability	No blank screen or unhandled error under network failure.	TC-ERR-01	Fault-injection test
FR-4.2	The system shall allow administrators to update flood polygons and checklist content via a versioned config file without a code redeploy.	Should	Ops / DRR team	New config version activates ≤1 min after publish.	TC-CFG-01	Config deployment test

4. Non-Functional Requirements

ID	Requirement	Metric / Target	Test Case	Verification
NFR-1	First Contentful Paint on low-end Android device (4× CPU throttle, Fast 3G: 1.6 Mbps / 150 ms RTT profile)	FCP ≤ 1.5 s; Lighthouse FCP score ≥ 0.90	TC-PERF-01	Lighthouse CI under 3G throttling
NFR-2	Offline capability: app shell, checklist, and last forecast cached via service worker	Core checklist renders fully offline; offline banner shown	TC-PWA-01	Service worker offline test (DevTools network offline)
NFR-3	Accessibility: WCAG 2.1 AA compliance	Contrast ≥ 4.5:1; keyboard navigable; screen-reader labels on all controls	TC-A11Y-01	axe-core audit + manual review
NFR-4	Payload: polygon and risk logic within budget	≤ 200 KB compressed on first visit	TC-PERF-02	Bundle-size CI gate
NFR-5	Compatibility matrix	Chrome/Edge ≥ 110, Safari ≥ 15, Firefox ≥ 110, KaiOS 2.5; feature-detection fallback	TC-COMP-01	Cross-browser matrix test
NFR-6	Privacy: no personal data leaves the device	Zero network calls containing user coordinates	TC-PRIV-01	Network traffic inspection

5. Requirements Traceability Summary

Each user need traces to at least one functional requirement and one or more test cases. Verification of bidirectional traceability (no orphan requirements, no orphan tests) is part of the review checklist before sign-off.

User Need / Source	Requirements	Test Cases
Persona A & B (location & checklist)	FR-1.1, FR-1.2, FR-3.2, FR-3.3	TC-GEO-01/02, TC-CHK-02, TC-STO-01
DRR Guidelines	FR-2.1	TC-SPAT-01
WMO / KMD Warnings	FR-2.2, FR-2.3, FR-2.4	TC-API-01, TC-ALGO-01/02
El Niño Action Plan	FR-3.1	TC-CHK-01
Persona C (Schools/Admin)	FR-3.4	TC-PRN-01

Reliability / Ops	FR-4.1, FR-4.2	TC-ERR-01, TC-CFG-01
Low-bandwidth need	NFR-1, NFR-2, NFR-4	TC-PERF-01/02, TC-PWA-01
FM-3 Mitigation	NFR-2	TC-PWA-01

6. Change Control

Flood polygons, risk thresholds, and checklist content are versioned in a configuration repository. Changes follow a two-person review (developer 1 + developer 2) and are deployed independently of application code. The active config version shall be displayed in the app footer for auditability.

7. Approval

Role	Name	Signature	Date
Dev 1	Lord_LLM		
Dev 2	KMN		